

Zoology Answer Keys

Every outstanding Zoology assignment, one section each. **The blue block is what goes on his paper**, sized to fit the box. Grey text underneath is context for you, not for copying.

ZOOLOGY · 15 POINTS

Classification Notes and Venn Diagram

This is Mrs. Finch's fill-in notetaking guide for the classification unit in Zoology: a warm-up plus ten numbered blanks about how scientists name and sort living things. Her header on the guide says the notes are for his own use and do NOT get turned in. The Classroom assignment it belongs to is titled W-Classification Notes and Venn Diagram, 15 points, due Sep 16, so a Venn diagram goes with it, but the guide itself does not describe that part.

1 Warm-up question (not numbered)

If you would have been an early scientist or settler, what is the one food, plant, or animal that you would have invented or discovered? And why?

His to supply. His own pick. Name one real food, plant, or animal, then one sentence on why it would have mattered to people. Two sentences is enough.

- His own answer
- Two sentences

BACKGROUND, FOR YOU

This is a hook question with no right answer. Anything real works: potatoes, chickens, coffee, honeybees, corn. If he stalls, ask him what he would not want to live without, and write that down.

Her header says: fill in, keep for your own notes, you do NOT need to turn in. So the guide itself is not collected. Worth knowing there is a conflict here: Classroom still lists a 15 point assignment called W-Classification Notes and Venn Diagram as missing. Ask her what she actually wants handed in.

2 1. The science of naming and classifying organisms

The science of naming and classifying organisms is called:

WRITE THIS · 1 WORDS

Taxonomy

- One word

BACKGROUND, FOR YOU

Taxonomy means sorting living things into groups and giving each group a name. A scientist who does this work is a taxonomist.

Some books call the same field systematics. Taxonomy is the word her blank is looking for.

3 2. How long ago Aristotle organized a system

Aristotle organized a system of classification, how long ago?

WRITE THIS · 4 WORDS

About 2,000 years ago

- Round number

BACKGROUND, FOR YOU

Almost every biology textbook rounds this to about 2,000 years ago, and that is what her blank wants. The real figure is bigger: Aristotle lived from about 384 to 322 BC, so roughly 2,350 years.

If she asks what his system was: he grouped animals by where they lived, land, water and air.

4 3. Who developed binomial nomenclature

The system of binomial nomenclature (we use now) was developed by:

WRITE THIS · 5 WORDS

Carolus Linnaeus (or Carl Linnaeus)

- Name to spell

BACKGROUND, FOR YOU

Binomial nomenclature just means two-name naming. Bi is two, nomen is name. Every organism gets a two-word name.

He was a Swedish scientist who published the system in the 1700s, and he is often called the father of taxonomy. Linnaeus is the part worth practicing to spell.

5 4. First letter of the genus

The first letter of the genus is always:

WRITE THIS · 4 WORDS

Capitalized (a capital letter)

- One word

BACKGROUND, FOR YOU

The genus is the first word of the two-word name, and it is the broader group.

This rule never changes, even in the middle of a sentence.

6 5. First letter of the species

The first letter of the species is always:

WRITE THIS · 4 WORDS

Lowercase (a small letter)

- One word
- Pairs with 4

BACKGROUND, FOR YOU

The species word is never capitalized, even when it is named after a person or a place.

Both words are italicized when typed and underlined when written by hand. Alex writes by hand, so he should underline both words.

7 6. The scientific name for humans

The scientific name for humans is correctly written as:

WRITE THIS · 2 WORDS

Homo sapiens

- Capitals matter
- Underline both words

BACKGROUND, FOR YOU

Capital H on Homo, lowercase s on sapiens. Underline both words when writing by hand. Typed, they would be in italics.

This question is really questions 4 and 5 combined. The common mistake is writing homo Sapiens or Homo Sapiens.

8 7. The language scientific names are written in

Scientific names are written in the language of ___ and give biologists a common way of communicating, regardless of the language they speak.

WRITE THIS · 1 WORDS

Latin

- One word

BACKGROUND, FOR YOU

Latin is not anybody's everyday language anymore, so the words do not drift or change meaning. That stability is the whole point.

A scientist in Japan and a scientist in Brazil both write Homo sapiens, so nobody has to translate. Some names use Greek word parts, but the convention is still called Latin.

9 8. The two parts of a scientific name

What 2 parts make up a Scientific Name?

WRITE THIS · 8 WORDS

Genus name and species name, in that order.

- Order matters

BACKGROUND, FOR YOU

Genus is the wider group that several close relatives share. Species is the one specific kind inside that group.

Dogs are *Canis familiaris* and wolves are *Canis lupus*. Same genus, *Canis*, so they are close relatives. Different species.

10 9. The seven categories in order (a through g)

What are the seven categories of classification in their proper order? (a through g)

WRITE THIS · 14 WORDS

a. Kingdom b. Phylum c. Class d. Order e. Family f. Genus g. Species

- Memory trick
- Order matters

BACKGROUND, FOR YOU

Each step down is narrower. Kingdom is the biggest bucket with millions of organisms. Species is the smallest, just one kind.

Memory trick, for his head and not for the paper: King Philip Came Over For Good Spaghetti. The first letters match: King (Kingdom), Philip (Phylum), Came (Class), Over (Order), For (Family), Good (Genus), Spaghetti (Species). Any sentence with those seven letters works, and one he makes up himself sticks better.

11 10. The definition being described

A biological ___ is a group of actually or potentially interbreeding natural populations that are reproductively isolated from other such groups.

WRITE THIS · 1 WORDS

Species

- One word
- Ties back to 8

BACKGROUND, FOR YOU

Plain-language version: a species is a group that can breed with each other and produce young that can also have young of their own, and that does not breed with outside groups.

Reproductively isolated means kept separate so they do not interbreed, whether by distance, timing, behavior, or biology.

12 Venn diagram (the other half of the assignment)

Compare two animals on a Venn diagram.

His to supply. Two overlapping circles. Left circle: traits only the first animal has. Right circle: traits only the second has. Middle: traits both share. He fills it in once he knows which two animals.

- His own answer
- Short phrases

BACKGROUND, FOR YOU

What is confirmed: the Classroom assignment is titled W-Classification Notes and Venn Diagram, 15 points, due Sep 16.

What is not confirmed: which two animals get compared. The notetaking guide file has the ten questions and nothing else, so the Venn instructions are not in anything we can read. He or Jonathan needs the Classroom post itself.

A Venn diagram is two overlapping circles. Anything true of only one animal goes in that animal's own part. Anything true of both goes in the middle where the circles cross.

Gap. The Venn diagram compares two stuffed animals and includes a Stuffed Animal Relay Race, per Finch's Classroom post. It is an in-class activity, so it cannot be completed at home.

Gap. Her guide says these notes are NOT collected, yet Classroom still lists this as a missing 15-point assignment. Worth asking her what she actually wants turned in.

Species Drawing, ID #1

This is a drawing assignment, not a writing one. Alex draws the eight animals from Mrs. Finch's "1 category names words" list and writes the category word and the common animal name under each drawing, on Google Slides or on paper.

1 How to set it up (read this first)

His to supply. Nothing goes on the paper for this part. Make eight boxes, one for each animal. Drawing on top, label underneath.

- Box 1 Bovine
- Box 2 Ovine
- Box 3 Porcine
- Box 4 Aves/Poultry
- Box 5 Equine
- Box 6 Caprine
- Box 7 Feline
- Box 8 Canine
- Box 9 Breed example (optional)

BACKGROUND, FOR YOU

The whole assignment is 15 points (that number comes from the Classroom listing, not from the doc). Nothing in the doc or in Finch's instructions asks for artistic quality, so a clear simple drawing with the correct label is the target.

Stick figures with the right features count. A cow needs to read as a cow, not win an art prize.

2 1. Bovine

Draw cattle, then write the label under the drawing.

His to supply. Bovine = Cattle (beef/dairy)

- Big rectangle body, four straight legs
- Large square head with a broad flat nose
- Two short horns curving up from the top of the head
- Two small ears sticking out sideways
- Thin tail with a tuft at the end
- Optional: a few patches on the body for a dairy cow

BACKGROUND, FOR YOU

Bovine is the category word for cattle. Both beef cattle and dairy cattle count, so one cow covers it.

The giveaway shapes are the boxy body and the two short curved horns. Those two things are what makes it read as cattle and not a horse.

3 2. Ovine

Draw a sheep, then write the label under the drawing.

His to supply. Ovine = Sheep

- Bumpy cloud shape for the woolly body
- Small narrow face, often drawn darker than the wool
- Two ears pointing down or out to the sides
- Four short thin legs
- Short tail, no horns needed

BACKGROUND, FOR YOU

Ovine is the category word for sheep.

Easiest animal on the list. A fluffy cloud shape with a small face is instantly readable as a sheep.

4 3. Porcine

Draw a pig, then write the label under the drawing.

His to supply. Porcine = Swine (pigs/hogs)

- Fat oval barrel body
- Round flat snout circle with two dots for nostrils
- Two triangle ears flopping forward
- Short stubby legs
- Curly pigtail, like a little spring

BACKGROUND, FOR YOU

Porcine is the category word for swine. The doc lists pigs and hogs together, so one pig covers it.

Write the label as swine, not just pig, since swine is the word the doc uses.

5 4. Aves / Poultry

Draw a chicken or a turkey, then write the label under the drawing.

His to supply. Aves/Poultry = Chickens/Turkeys

- Oval body sitting fairly upright
- Small round head on a short neck
- Zigzag comb on top of the head
- Small triangle beak, with a wattle hanging below it
- Two thin legs, three toes pointing forward on each
- A few curved tail feathers sweeping up at the back

BACKGROUND, FOR YOU

Aves is the category word for birds. Poultry is the farm version of that word, meaning birds raised for meat and eggs. One bird is enough. A chicken is much easier to draw than a turkey.

6 5. Equine

Draw a horse, then write the label under the drawing.

His to supply. Equine = Horses

- Long rectangle body
- Long neck angling up from the shoulders
- Long narrow head with a small muzzle
- Mane as a jagged line running down the back of the neck
- Long flowing tail
- Four long thin legs ending in small hooves

BACKGROUND, FOR YOU

Equine is the category word for horses.

The long neck and the mane are what separate a horse from the cattle drawing. Keep the legs long and thin.

7 6. Caprine

Draw a goat, then write the label under the drawing.

His to supply. Caprine = Goats

- Smooth rectangle body, not fluffy
- Narrow head with a small beard hanging under the chin
- Two horns sweeping back over the neck
- Ears pointing out to the sides
- Short tail pointing up, not down
- Four thin legs

BACKGROUND, FOR YOU

Caprine is the category word for goats.

Look back at the sheep in box 2 while drawing this one, so the difference is obvious on the page. Smooth coat, beard, horns sweeping back, tail pointing up.

8 7. Feline

Draw a cat, then write the label under the drawing.

His to supply. Feline = Cats

- Oval body, sitting or standing
- Round head with two triangle ears on top
- Three whiskers drawn on each side of the nose
- Long tail curving up or curled around the body
- Four short legs with small paws

BACKGROUND, FOR YOU

Feline is the category word for cats.

Triangle ears and whiskers are the whole job here.

9 8. Canine

Draw a dog, then write the label under the drawing.

His to supply. Canine = Dogs

- Oval body, standing side on
- Head with a long rectangle snout sticking forward
- Two floppy ears hanging down, or two pointed ears standing up
- Tail sticking up or out behind
- Four legs with small paws
- Optional: a collar line around the neck

BACKGROUND, FOR YOU

Canine is the category word for dogs.

Drawn next to the cat, a dog needs a longer snout and floppy or larger ears so the two are not confused.

10 9. Breed, the Angus example (optional extra)

Optional. Draw beef cattle a second time and write both label lines under it.

His to supply. Breed is more specific than the Genus/Species. Bos taurus (Beef cattle) - Angus (Breed of Beef cattle)

- Same cattle shape as box 1: rectangle body, four legs, boxy head
- Color it in solid, or leave it plain
- Write both lines of the label under it, the general name first and the breed second

BACKGROUND, FOR YOU

Read this before deciding. The doc's list has eight animals. Breed is a separate section of the same doc, and the doc does not say it needs its own drawing. Treat it as optional.

It is quick if he does it: the drawing is the same cattle shape as box 1, and the two label lines are copied straight from the doc.

Gap. Her doc gives a scientific genus and species for only ONE animal, Bos taurus for cattle. The other seven have no scientific name anywhere in her material, so do not invent them.

Gap. Aves/Poultry is listed as chickens AND turkeys. One drawing is the reasonable read, but worth confirming.

"What is Zoology?" Poster

This is the 15 point poster for Mrs. Finch's Zoology class, built from her "1 What is Zoology NT Guide" vocabulary sheet. Her guide gives seven definitions with the term blanked out, so the poster's job is to show the seven terms with their meanings, plus a section on careers in zoology, laid out so it reads clearly from a few feet away.

1 Poster title (letter this at the top)

Where do you want the title to sit, across the top or down one side?

WRITE THIS · 3 WORDS

What is Zoology?

- Title
- Poster

BACKGROUND, FOR YOU

Keep the title the biggest text on the page. One short line, centered or across the top.

If he wants a subtitle, "7 words every zoologist knows" works, but it is optional and the guide does not ask for one.

2 Term 1: Zoology

Her definition has five parts. Which one would you most want to study?

WRITE THIS · 27 WORDS

ZOOLOGY The science of animals. It studies how animals behave, how their bodies are built, how their bodies work, how they are grouped, and where they live.

- Vocabulary
- Term 1 of 7

BACKGROUND, FOR YOU

Her definition word for word: "the scientific study of the behavior, structure, physiology, classification, and distribution of animals."

That definition has FIVE parts, and she has quizzed them as five: behavior, structure, physiology, classification, distribution. The line above keeps all five in order, which is why "built" and "work" are split into two clauses instead of one.

3 Term 2: Biology

If biology is all life, what else besides animals fits inside it?

WRITE THIS · 15 WORDS

BIOLOGY The study of life. Zoology is one branch of biology, the branch about animals.

- Vocabulary
- Term 2 of 7

BACKGROUND, FOR YOU

Her definition word for word: "the study of life."

The second sentence is not in her guide. It is there because it shows how the first two terms connect, which makes the poster hold together. Drop it if he wants less to write.

4 Term 3: Wildlife

Is a zoo lion wildlife? Good one to argue about, the species is, the individual animal is living in a zoo.

WRITE THIS · 23 WORDS

WILDLIFE Animals and plants that are not domesticated. They live on their own in nature instead of being raised or kept by people.

- Vocabulary
- Term 3 of 7

BACKGROUND, FOR YOU

Her definition word for word: "non-domesticated animals (& plants)." The plants are in parentheses in her guide, so keep them in.

Domesticated means tamed and raised by people over many generations. Cows, dogs, and chickens are domesticated. Deer and hawks are not.

5 Term 4: Aquaculture

What is the difference between aquaculture and just fishing?

WRITE THIS · 21 WORDS

AQUACULTURE Raising fish in a controlled water environment, like a tank, pond, or pen, instead of catching them in the wild.

- Vocabulary
- Term 4 of 7

BACKGROUND, FOR YOU

Her definition word for word: "the growing of fish in a controlled water environment."

Think of it as farming, but with fish. Aqua means water, culture means growing something.

6 Term 5: Zoo

Her definition lists three reasons zoos exist. Can you name all three without looking?

WRITE THIS · 18 WORDS

ZOO A place that keeps a collection of wild animals for study, conservation, or display to the public.

- Vocabulary
- Term 5 of 7

BACKGROUND, FOR YOU

Her definition word for word: "establishment that maintains a collection of wild animals for study, conservation, or display to the public."

Conservation means protecting animals and their habitats so a species does not die out.

7 Term 6: Veterinary

Veterinary or veterinarian, which one is the job title and which one describes the work?

WRITE THIS · 18 WORDS

VETERINARY Having to do with the diseases, injuries, and treatment of animals. A veterinarian is an animal doctor.

- Vocabulary
- Term 6 of 7

BACKGROUND, FOR YOU

Her definition word for word: "relating to the diseases, injuries, and treatment of animals."

The word she blanked out is the describing word "veterinary," not the job title "veterinarian." Veterinary science, veterinary clinic. Worth saying once so he knows which word fills her blank.

8 Term 7: Marine biology

Marine biology and zoology overlap. What kind of animal would both of them study?

WRITE THIS · 11 WORDS

MARINE BIOLOGY The scientific study of living things in the sea.

- Vocabulary
- Term 7 of 7

BACKGROUND, FOR YOU

Her definition word for word: "the scientific study of organisms in the sea."

Organism just means a living thing. Marine comes from mare, the Latin word for sea.

9 Careers in zoology

Pick the one you would actually want. That is the one to draw.

WRITE THIS · 12 WORDS

CAREERS IN ZOOLOGY Zoologist. Zookeeper. Veterinarian. Marine biologist. Wildlife biologist. Fish farmer.

- Careers
- Own knowledge, not the guide

BACKGROUND, FOR YOU

Her guide lists "Careers involved in Zoology" as a topic but does not name a single career, so this list comes from general knowledge, not from her sheet. If she has a list of her own, use hers instead.

These six are chosen so each one ties back to one of her seven terms: zoologist to zoology, zookeeper to zoo, veterinarian to veterinary, marine biologist to marine biology, wildlife biologist to wildlife, fish farmer to aquaculture. If she asks why these six, that is the answer.

10 Layout and drawings

Do you want to rule the boxes yourself or should I pencil them in for you?

His to supply. Alex supplies this part. The layout and the drawings are his own. Plan: title across the top, the seven terms in boxes below it, careers in a box at the bottom, drawings in the leftover space.

- Student's own work
- Layout

BACKGROUND, FOR YOU

The layout is his to decide and the drawings have to be his own, so this section is a plan, not an answer to copy.

Low hand fatigue version: rule the paper into 3 boxes across and 4 down, which is 12 boxes. The top row of 3 becomes one wide title banner. That leaves 9 boxes for 7 terms plus careers, with 1 box spare for a drawing. Nothing to measure, and it comes out even.

Gap. Her guide lists 'Careers involved in Zoology' as a topic but names no careers, so the career list here is general knowledge, not hers.

Gap. Nothing specifies poster size, material, or whether drawings are required. Treat the layout suggestion as a suggestion.